Anexo 7: Reporte de Procesamiento de Líneas Base GNSS – Vértice gps1

Corresponde al informe técnico de procesamiento de líneas base GNSS para el punto gps1 determinado desde las estaciones de control BUEN y CALI, detallando los parámetros de cálculo, calidad de geometría satelital (DOP) y resolución de ambigüedades en fase fija.

Leica Geosystems AG
Heinrich Wild Strasse
CH-9435 Heerbrugg
St. Gallen, Switzerland

Phone: + 41 71 727 3131
Fax: + 41 71 727 4674

- when it has to be right

GNSS Processing Report - Summary

Report created: 06/10/2026 19:05:49

Project Details

General		Customer Details	
Project Name:	Daniel Cortes	Customer Name:	-
Owner:	-	Contact Person:	-
Lead Surveyor:	-	Number:	-
Date Created:	06/10/2026 18:18:57	Email:	-
Last Accessed:	06/10/2026 18:18:57	Skype:	-
Application Software:	Infinity 3.0	Website:	-

Master Coordinate System	
Coordinate System Name:	MAGNA-SIRGAS_Origen-Nacional
Transformation Type:	Classical 3D
Residual Distribution:	None
Ellipsoid:	GRS 1980
Projection Type:	Transverse Mercator
Geoid Model:	Geocol2004
CSCS Model:	-

Path:	C:\Users\usuario\Documents\Leica Geosystems\Infinity\Projects\Daniel Cortes\Daniel Cortes.iprj
Size:	57,2 MB
Comments:	-

Baseline BUEN - gps1

Processing Parameters (04/10/2026 11:00:27 - 04/10/2026 15:04:39)

Data	Selected	Used	Comments
Cut-Off Angle:	10°	10°	
Frequency:	Automatic	L1/L2	
Sampling Rate:	Use All	15,00 sec	
Satellite System:	GPS/GLONASS/Galileo/Beidou	GPS/GLONASS	
Ephemeris Type:	Precise	Precise	
Antenna Calibration Set:	NGS Absolute	NGS Absolute	

Processing Strategy

Solution Type:	Phase Fixed	Phase Fixed
Solution Optimisation:	Automatic	Iono Minimised
Frequency to use in Iono Minimised:	Automatic	L1/L2
Tropospheric Model:	VMF with GPT2 model	VMF with GPT2 model
Ionospheric Model:	Automatic	Computed
Allow Widelane Fix:	Automatic	Automatic

General Settings

Min. Distance for Iono Minimised:	15 km
Possible Ambiguities Fix up to:	300 km
Min. Duration for Float Solution (static):	00:05:00

Results Baseline: BUEN - gps1

Acquisition

Start Time - End Time:	04/10/2026 11:00:27 - 04/10/2026 15:04:27
Duration:	04:04:00

Antennas

	Reference - BUEN	Rover - gps1

Receiver Name / SN:	TRIMBLE NETR9 / 5134K78077	V30 Plus / 11662288
Antenna Name / SN:	TRM59800.00 SCIS / 5138354298	HITV30PLUS / -
Carrier Offset:	-	-
Height Reading:	0,1240 m	1,9196 m
Antenna Height:	0,1240 m	1,9196 m

Coordinates

	Reference - BUEN	Rover - gps1		Reference - BUEN	Rover - gps1
Point Role:	Control	Fixed PP			
WGS84 Latitude:	3° 52' 55,29063" N	3° 29' 05,30972" N	Easting:	4.554.573,0135 m	4.644.746,3911 m
WGS84 Longitude:	77° 00' 37,51018" W	76° 11' 53,04759" W	Northing:	1.988.021,5479 m	1.943.679,3026 m
WGS84 Ellip. Height:	57,7645 m	1.536,1960 m	Ortho. Height:	39,2398 m	1.503,9926 m
WGS84 Cartesian X:	1.430.383,9669 m	1.519.178,9090 m			
WGS84 Cartesian Y:	-6.200.818,1398 m	-6.184.088,7686 m			
WGS84 Cartesian Z:	428.934,2670 m	385.190,7603 m			

Baseline Vector and Quality - WGS84

ΔLatitude:	-0° 23' 49,98090"	SD ΔLatitude:	0,0011 m
ΔLongitude:	0° 48' 44,46260"	SD ΔLongitude:	0,0014 m
ΔHeight:	1.478,4315 m	SD ΔHeight:	0,0035 m
ΔX:	88.794,9421 m	SD ΔX:	0,0016 m
ΔY:	16.729,3712 m	SD ΔY:	0,0035 m
ΔZ:	-43.743,5067 m	SD ΔZ:	0,0011 m
Slope Dist.:	100.388,7841 m	SD Slope Dist.:	0,0013 m

M0:	1,3772 m	CQ 1D:	0,0035 m
Q11:	0,00000129	CQ 2D:	0,0018 m
Q12:	-0,00000108	CQ 3D:	0,0040 m
Q22:	0,00000633		
Q13:	0,00000016		
Q23:	-0,00000052		
Q33:	0,00000067		

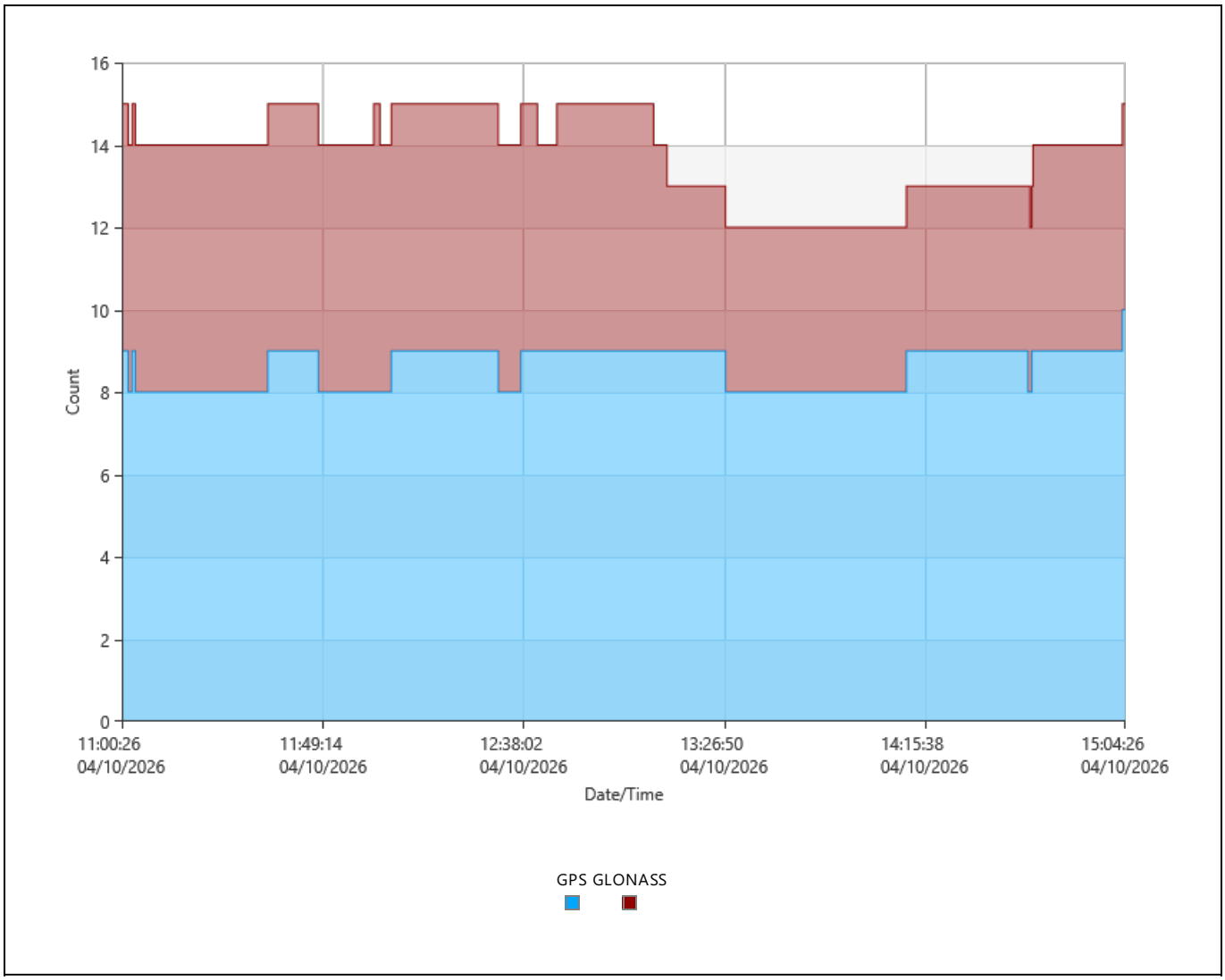
Frequency:	L1/L2	GDOP:	1,8 - 3,5	GPS SVs:	8/10
Solution Optimisation:	Iono Minimised	PDOP:	1,4 - 2,7	GLONASS SVs:	6/7
Solution Type:	Phase Fixed	HDOP:	0,7 - 1,0	Beidou SVs:	-
		VDOP:	1,1 - 2,5	Galileo SVs:	-
				QZSS SVs:	-

Ephemeris Type:	
GPS	Precise
GLONASS	Precise

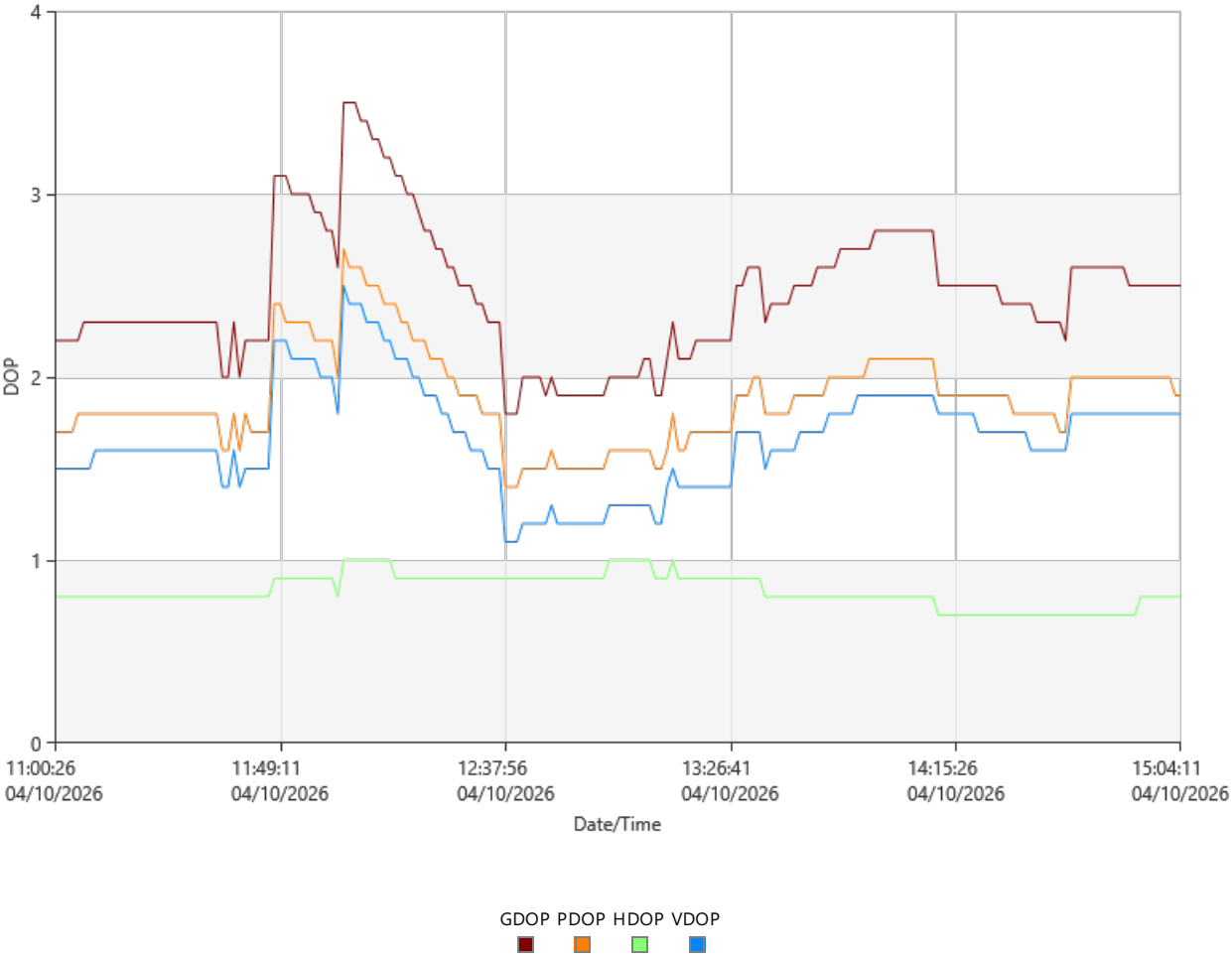
Processing Info (04/10/2026 11:00:27 - 04/10/2026 15:04:39)

Processed Date/Time:	06/10/2026 19:00:33

SVs Tracked


16
14
12
10
8
6
4
2
0
Count
11:00:26
04/10/2026
11:49:14
04/10/2026
12:38:02
04/10/2026
13:26:50
04/10/2026
14:15:38
04/10/2026
15:04:26
04/10/2026
Date/Time
GPS GLONASS

DOP



Ambiguity Statistics

Number of Ambiguities	GPS	GLONASS
Fixed	34	15
Total	43	21
Independently fixed	375	307
Possible independently fixed	838	838

Average time between independent fixes: 00:01:15

% of Epochs	GPS		GLONASS	
	L1 [%]	L2 [%]	L1 [%]	L2 [%]
Fixed	83,30	85,19	98,18	87,93
Not fixed	1,27	1,22	1,82	1,79
Not fixed - contradiction	15,43	13,59	0,00	10,28
Not fixed - missing phase	0,00	0,00	0,00	0,00

Status	From Epoch	To Epoch	Duration
Fixed	04/10/2026 11:00:27	04/10/2026 15:04:27	04:04:00

Baseline CALI - gps1

Processing Parameters (04/10/2026 11:00:27 - 04/10/2026 15:04:39)

Data	Selected	Used	Comments
Cut-Off Angle:	10°	10°	
Frequency:	Automatic	L1/L2/L5	
Sampling Rate:	Use All	15,00 sec	
Satellite System:	GPS/GLONASS/Galileo/Beidou	GPS/GLONASS	
Ephemeris Type:	Precise	Precise	

Antenna Calibration Set:	NGS Absolute	NGS Absolute

Processing Strategy

Solution Type:	Phase Fixed	Phase Fixed
Solution Optimisation:	Automatic	Iono Minimised
Frequency to use in Iono Minimised:	Automatic	L1/L2/L5
Tropospheric Model:	VMF with GPT2 model	VMF with GPT2 model
Ionospheric Model:	Automatic	Computed
Allow Widelane Fix:	Automatic	Automatic

General Settings

Min. Distance for Iono Minimised:	15 km
Possible Ambiguities Fix up to:	300 km
Min. Duration for Float Solution (static):	00:05:00

Results Baseline: CALI - gps1

Acquisition

Start Time - End Time:	04/10/2026 11:00:27 - 04/10/2026 15:04:27
Duration:	04:04:00

Antennas

	Reference - CALI	Rover - gps1
Receiver Name / SN:	TRIMBLE NETR9 / 5145K79436	V30 Plus / 11662288
Antenna Name / SN:	TRM59800.00 SCIS / 5134354245	HITV30PLUS / -
Carrier Offset:	-	-
Height Reading:	0,1400 m	1,9196 m
Antenna Height:	0,1400 m	1,9196 m

Coordinates

	Reference - CALI	Rover - gps1		Reference - CALI	Rover - gps1
Point Role:	Control	Fixed PP			
WGS84 Latitude:	3° 22' 32,84048" N	3° 29' 05,31530" N	Easting:	4.607.499,9398 m	4.644.746,3286 m
WGS84 Longitude:	76° 31' 57,23035" W	76° 11' 53,04963" W	Northing:	1.931.743,8449 m	1.943.679,4743 m
WGS84 Ellip. Height:	1.027,4846 m	1.536,0807 m	Ortho. Height:	998,5097 m	1.503,8774 m
WGS84 Cartesian X:	1.483.100,0097 m	1.519.178,8179 m			
WGS84 Cartesian Y:	-6.193.060,1469 m	-6.184.088,6619 m			
WGS84 Cartesian Z:	373.124,3315 m	385.190,9244 m			

Baseline Vector and Quality - WGS84

ΔLatitude:	0° 06' 32,47483"	SD ΔLatitude:	0,0003 m
ΔLongitude:	0° 20' 04,18072"	SD ΔLongitude:	0,0003 m
ΔHeight:	508,5962 m	SD ΔHeight:	0,0008 m
ΔX:	36.078,8082 m	SD ΔX:	0,0004 m
ΔY:	8.971,4850 m	SD ΔY:	0,0007 m
ΔZ:	12.066,5928 m	SD ΔZ:	0,0003 m
Slope Dist.:	39.086,7063 m	SD Slope Dist.:	0,0003 m

M0:	0,3762 m	CQ 1D:	0,0008 m
Q11:	0,00000108	CQ 2D:	0,0004 m
Q12:	-0,00000090	CQ 3D:	0,0009 m
Q22:	0,00000372		
Q13:	0,00000004		
Q23:	-0,00000026		
Q33:	0,00000047		

Frequency:	L1/L2/L5	GDOP:	1,5 - 2,2	GPS SVs:	9/10
Solution Optimisation:	Iono Minimised	PDOP:	1,2 - 1,7	GLONASS SVs:	7/7
Solution Type:	Phase Fixed	HDOP:	0,6 - 0,8	Beidou SVs:	-

		VDOP:	1,0 - 1,5	Galileo SVs:	-
				QZSS SVs:	-

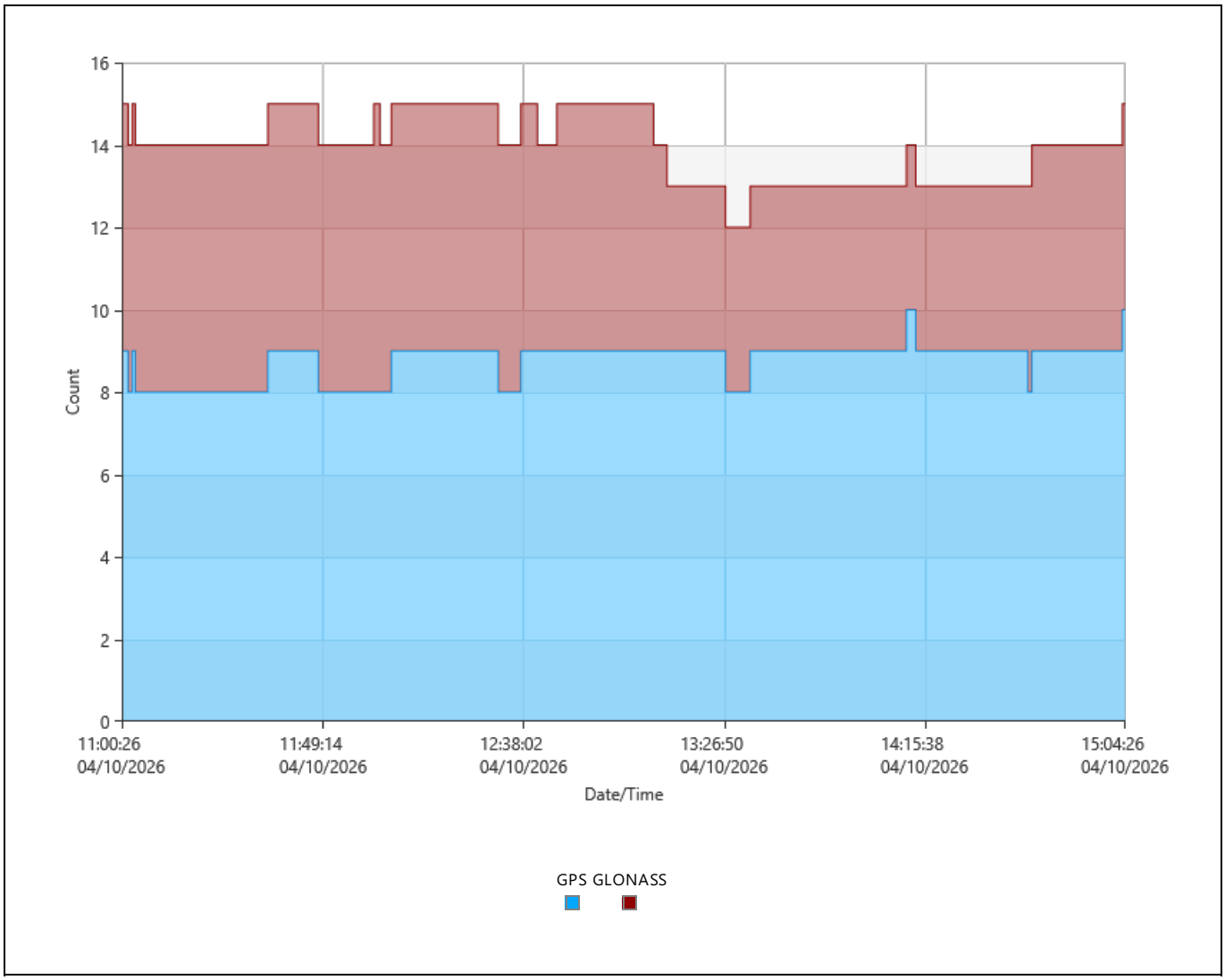
Ephemeris Type:

GPS	Precise
GLONASS	Precise

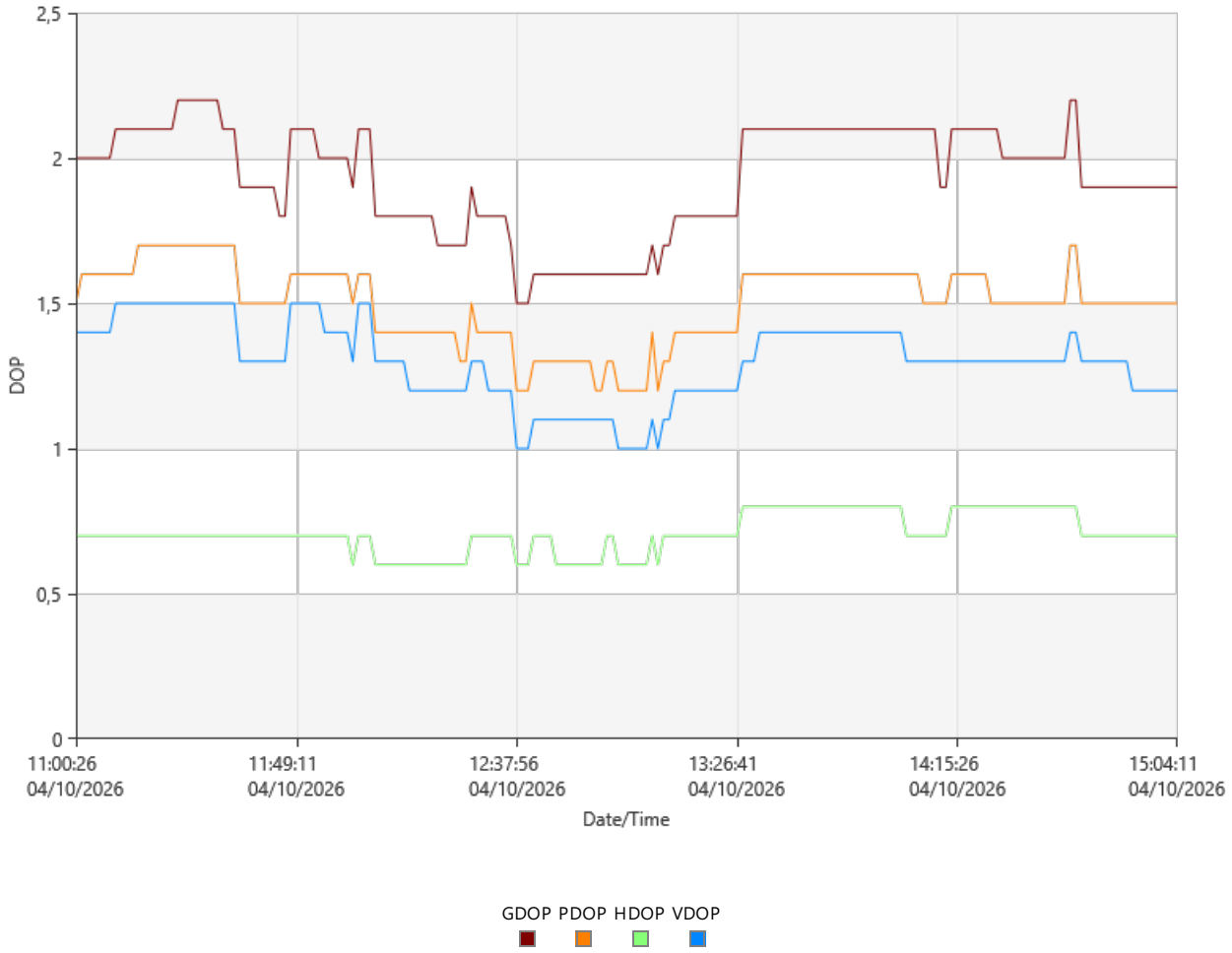
Processing Info (04/10/2026 11:00:27 - 04/10/2026 15:04:39)

Processed Date/Time:	06/10/2026 19:00:33

SVs Tracked



DOP



Ambiguity Statistics

Number of Ambiguities	GPS	GLONASS
Fixed	41	20
Total	52	24
Independently fixed	703	701
Possible independently fixed	801	801

Average time between independent fixes: 00:01:45

% of Epochs	GPS			GLONASS	
	L1 [%]	L2 [%]	L5 [%]	L1 [%]	L2 [%]
Fixed	95,42	95,45	93,96	98,10	98,00
Not fixed	0,30	0,27	0,06	1,90	2,00
Not fixed - contradiction	4,28	4,29	5,97	0,00	0,00
Not fixed - missing phase	0,00	0,00	0,00	0,00	0,00

Status	From Epoch	To Epoch	Duration
Fixed	04/10/2026 11:00:27	04/10/2026 15:04:27	04:04:00